A Project Report

On

Loan Approval Prediction System

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted to

Department of Computer Application

Xavier International College

Boudha, Bhaktapur

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Tribhuvan University

Faculty of Humanities and Social Sciences

Xavier International College

Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **Elizabeth Subba, Sneha Shrestha** entitled "**Loan Approval Prediction System**" in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

_____

Signature

Roshan Jha

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LETTER OF APPROVAL

This is to certify that this project prepared by Elizabeth Subba and Sneha Shrestha entitled **"Loan Approval Prediction System"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

_____ **SIGNATURE of Supervisor** Roshan Jha BCA Department Xavier International College	_____ **SIGNATURE of Coordinator**
_____ **SIGNATURE of Internal Examiner**	_____ **SIGNATURE of External Examiner**

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This project has been a wonderful experience where we have learnt and experienced many beneficial things.

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Abstract

Loan prediction is a critical area in banking and insurance, where data analysis plays a pivotal role to understand patterns in the data. Loan prediction involves the use of machine learning methods such as Random Forest, implemented in Python with tools like Google Colab.

The project “Loan Approval Prediction System” focuses on using machine learning, specifically, Random Forest to predict loans. The machine learning model is created using Python and implemented through a Flask API. The frontend of the project is designed using Next.js, while the backend is developed using Node.js. The system delivers accurate results with reasonable loss during training and validation. The results suggest that the model has 82.5% accuracy. The system can improve this work by focusing on achieving even higher accuracy in the future.

***Key Words**: Loan, Prediction System, Random Forest*

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List of Abbreviation

ANN	: Artificial Neural Networks (ANN)
API	: Application Program Interface
CSS	: Cascading Style Sheet
GDRP	: General Data Protection Regulation
HTML	: Hyper Text Markup Language
I/O	: Input/Output
JS	: Java Script
PC	: Personal Computer
SQL	: Structured Query Language
SVM	: Support Vector Machine
UI	: User Interface
UML	: Unified Modeling Language
WSGI	:Web Server Gateway Interface

Chapter 1. Introduction

1.1. Introduction

With the increasing number of customers approaching banks and financial institutions for loans, the risk of loan defaulting has become a significant concern. Loan approval is a crucial decision for banks as it directly impacts their profitability and stability. To mitigate this risk, banks are turning to advanced technologies such as machine learning and computer science to predict the probability of defaulting for potential borrowers. By analyzing various factors, these technologies enable banks to assess an individual's capacity to repay a loan within the proposed terms. And, Laon Approval Prediction is a similar system that which perform predictive analysis to make informed decisions and minimize the chances of non-performing loans, ultimately safeguarding the stability of the banking sector and reducing the risk of bankruptcy.

The project focuses on designing an intuitive user interface that allows loan officers or financial institutions as well as general users to input applicant information, such as personal details, income, credit history, and employment status. This data serves as the input for the loan approval prediction algorithm, which analyzes various factors to generate a prediction of loan approval or rejection. The system aims to provide a clear and understandable prediction result, enabling loan officers to make well-informed decisions based on the algorithm's output. Additionally, the UI may include features such as data validation, error handling, and notifications to ensure data accuracy and enhance user experience throughout the loan application process.

The Loan Approval Prediction project not only benefits lending institutions but also applicants seeking loans. By leveraging historical loan data and advanced analytics, the system can assess loan applications objectively and fairly. This approach reduces bias and subjectivity in the decision-making process, providing applicants with a transparent and reliable loan approval prediction. Moreover, the project contributes to the overall efficiency of loan processing, potentially reducing turnaround time for loan approvals and enhancing customer satisfaction. The accurate prediction of loan approvals can assist applicants in making informed financial decisions and improving their chances of successful loan applications.

1.2. Problem Statement

The problem this project aims to address is the difficulty faced by applicant when issuing the loan. It also helps loan officer to find the person who is capable to pay loan. The loan approval process in banks and financial institutions is a critical decision-making task that involves assessing the creditworthiness and risk associated with potential borrowers. The increasing number of loan applicants further exacerbates the challenge of accurately and efficiently evaluating loan requests. The traditional approach heavily relies on manual assessment by loan officers, leading to inconsistencies and inefficiencies in the decision-making process. Additionally, the risk of approving loans to customers with a high probability of defaulting poses a significant threat to the financial stability of the institutions. So, introducing the loan approval prediction system can be a useful tool for loan approval for the developing country like Nepal.

1.3. Objectives

The objectives of Loan Approval Prediction System are:

i. To predict loan by using machine learning Random Forest Algorithm.

ii. To provide information about personal loan.

1.4. Scope and Limitation

The scope of a Loan Approval Prediction System is that it helps banks and lenders decide whether to approve or reject loan applications. It uses data to make predictions about whether a person is likely to repay a loan or not using Random Forest machine learning algorithm.

However, it also has limitations. It provides loan approval status based on general condition but not specific to the any banking particular banking rule. While it's a useful tool, it's not completely reliable, and lenders should also take other factors into account when deciding on loans.

1.5. Development Methodology

System development is a process through which a product will get completed or a product gets rid from any problem. Software development process is described as a number of phases, procedures and steps that gives the complete software. It follows series of steps

which is used for product progress. The development method followed in this project is waterfall model.

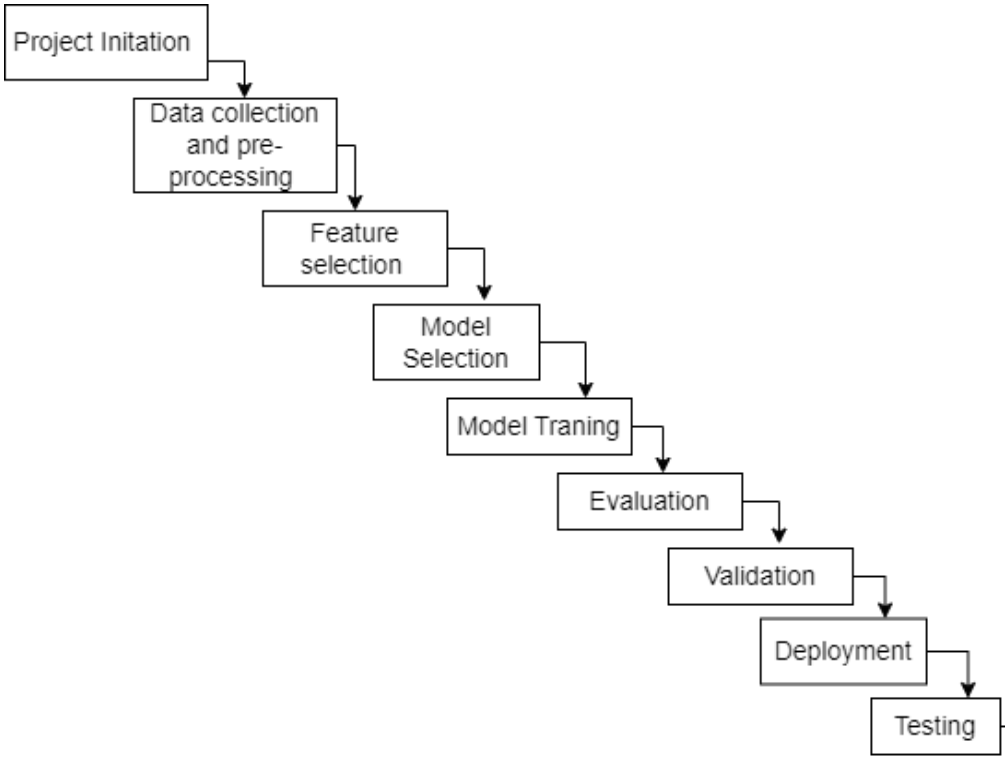


Figure 1.1 Waterfall Model of Loan Approval Prediction System

The development methodology for a Loan Prediction System typically involves several key steps:

i. Project Initiation: The project's goals, objectives, and scope are identified and gather initial requirements.

ii. ii. Data Collection and Preprocessing: Historical loan data was collected, and it was cleaned and preprocessed to eliminate inconsistencies and missing values. This step is crucial because the quality of data significantly affects the accuracy of predictions.

iii. Feature Selection: Relevant features (variables) were identified from the dataset, and new features were created if necessary. Feature selection was employed to enhance the model's performance and reduce complexity.

iv. Model Selection: The appropriate machine learning or statistical models for loan prediction were chosen. Common choices, depending on the complexity of the

problem, included logistic regression, decision trees, random forests, or neural networks.

v. Model Training: A portion of the data was used to train the selected model, and this involved adjusting model parameters to fit the data accurately.

vi. Model Evaluation: The model's performance was evaluated using metrics like accuracy, precision, recall, and F1-score.

vii. Validation: The model's hyperparameters were fine-tuned to optimize its performance.

viii. Deployment: Once a satisfactory model was achieved, it was integrated into the Loan Approval System. An API or user interface was created for interacting with the model.

ix. Testing: The entire system was thoroughly tested to ensure it functioned as expected and delivered accurate loan predictions.

x. Documentation and Testing: The entire development process was documented, including data sources, model architecture, and deployment details. A maintenance plan was implemented for regular updates and monitoring of the model's performance.

1.6. Report Organization

The report consists of four chapters. The “Introduction” is the first chapter which contains whole information about the project “Loan Approval Prediction System”. The introduction chapter contains the introduction, problem statement, objectives and scope of the project. The second chapter includes the background study and literature review. The chapter three is known as system analysis and design which include many sub chapters. The sub chapters inside the chapter three are: system analysis and system design which is further divided into sub topics. The system analysis includes requirement analysis (functional analysis and non-functional analysis), feasibility study (Technical feasibility, operational feasibility, economic feasibility and scheduling feasibility), object diagram and sequence diagram. The system design consists of state modeling, dynamic modeling and deployment modeling the project. The fourth chapter includes implementation and testing with result analysis. The last chapter contains two parts:

Conclusion and Future recommendation.

Chapter 2. Background Study and Literature Review

2.1. Background Study

In Nepal, the loan approval prediction process in banks and financial institutions traditionally involves manual assessment by loan officers. They evaluate loan applications based on various factors, including the applicant's credit history, income, collateral, and other relevant financial information. The decision-making process is subjective and heavily relies on the experience and judgment of the loan officers.

In recent years, there has been a growing interest in leveraging technology and data-driven approaches for loan approval prediction in Nepal. Some financial institutions have started exploring the use of machine learning and data analytics techniques to enhance the accuracy and efficiency of loan assessment. But, still there is not a platform for general people or applicant to predict either they are applicable for issuing the loan or not.

The "Loan Approval Prediction System" project has a primary objective: to bridge the existing gap in the loan approval process within Nepal's financial institutions. Traditional methods involve manual assessment by loan officers, which can be time-consuming, subjective, and reliant on human judgment.

The scope of the project encompasses the development of a robust machine learning model, specifically utilizing the Random Forest algorithm, implemented in Python and accessible through a Flask API. Random Forest is a versatile machine learning algorithm known for its ability to handle complex datasets and improve predictive accuracy by aggregating predictions from multiple decision trees. It works by creating a collection of decision trees and combining their results to make robust and accurate predictions in various applications, from classification to regression tasks.

The project's architecture includes Python, Flask API, Next.js, and Node.js as key components, forming a cohesive foundation for its development. Python is widely preferred for developing machine learning models due to its extensive libraries, simplicity, and rich ecosystem, making it an ideal language for data analysis, model training, and deployment. Flask API is a lightweight and flexible framework for building web APIs in Python, facilitating the creation of web-based applications with ease [1]. Next.js is a powerful JavaScript framework that simplifies server-side rendering and provides an efficient, optimized, and user-friendly approach for building modern web applications. Node.js,

which serves as a JavaScript runtime, is employed to efficiently handle data processing and manage server-side operations within the project [2].

2.2. Literature Review

With the increase in banking sector many people are applying for loans in bank. All these loans are not approvable. The main income of bank assets comes from gain earned from loans. The main objective of banks is to invest their assets in safe customers. Today many banks approve loan after many processes of verification and validation but still there is no surety that selected customer is safe or not. Therefore, the "Loan Approval Prediction System Using Machine Learning" by Associate Professor and Head of Department of Computer Science and Engineering from Hyderabad, India concluded it is important to apply various techniques in banking sector for selecting a customer who pays loan on time [3].

The paper with title "Bank Loan Approval Prediction Using Machine Learning Techniques" by Ndayisenga, Theoneste concluded that financial institutions should use machine learning techniques because it saves money and time for both sides. [4].

In depth analysis of credit scoring in various fields with research paper "Credit Scoring, Statistical Techniques and Evaluation Criteria" by Hussein A. Abdou and John Pointon in 2011 concluded that there is no single overall best classification technique for credit scoring models [5].

The top 10 important features from the models were selected and then used in the modeling process to test the stability of binary classifiers by comparing their performance on separate data in 2018 by authors of paper "Credit Risk Analysis Using Machine and Deep Learning Models" observe that the tree-based models are more stable than the models based on multilayer artificial neural networks. [6]

Mostafa Yousofi Tezerjan and Azizollah Memariani author of "A hybrid model for credit scoring in complex systems" concluded that using machine learning system will allow the lending bank to be informed of the outcome of the current situation and credit history of its customers, which constitutes the customer [7].

The article title "Loan default prediction using decision trees and random forest" concluded that "the Random Forest algorithm demonstrated superior performance compared to the Decision Tree algorithm, achieving significantly greater accuracy". [8].

Paper introduced by Lifeng Zhou and Hong Wang* in 2012 with title Loan Default Prediction on Large Imbalanced Data Using Random Forests shows that parallel random forests can greatly improve random forests' efficiency during the learning process. [9].

Svraka, and Nalica introduced the article "Bank Loan Prediction Using Machine Learning Techniques" which conclude models typically assign a numerical score (1 or 0), which represents the creditworthiness of the borrowers based on historical data [10].

The article purposed by D. J. HAND and W. E. HENLEY with the title "Statistical Classification Methods in Consumer Credit Scoring" concluded the statistical work in credit scoring and credit control to date has focused on the conceptually relatively straightforward aspect of constructing improved discriminating rules [11].

The paper "Secure Loan Prediction System Using Artificial Neural Network (ANN)" conclude that the use of a decision support system and the expert system has helped to solve major issues in the banking industry and the world at large and also increasing the accuracy of the decision [12].

To determine the maximum relevant features, i.e. the factors that have the most impact on the prediction outcome, various ML algorithms such as Random Forest, Support Vector Machine, K-Nearest Neighbor and Logistic Regression, were used. These mentioned algorithms are evaluated with the standard metrics and compared with each other. The random forest algorithm achieves better accuracy. This paragraph is the conclusion of the article "Comparative Analysis of Customer Loan Approval Prediction using Machine Learning Algorithms" [13].

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Chapter 3. System Analysis and Design.

3.1. System Analysis

3.1.1. Requirement Analysis

a. Functional Requirement

Functional requirements are product features or functions those developers must implement to enable users to accomplish their tasks. The functional requirement of our project:

i. Login/Register

Loan approval predictor allows the user to create a new account or log into an existing one.

ii. View Applicant

The admin of the Loan approval predictor can view the list of the applicant or the user how predict the loan approval using this system.

iii. Input loan specification

Loan approval predictor enables the user to input the specifications. So, that they can get information about their loan request approval.

iv. Predict Loan

The loan approval predictor informs about approval loan by using machine learning model.

v. Download Individual Prediction.

The loan approval predictor enables the user to download a report containing the predicted status and other details of their own.

vi. Download Prediction.

The loan approval predictor enables the admin to download a report containing the predicted price and other details of all users.

vii. Manage Blog.

The admin of the system can insert, update and delete the bog through the admin dashboard.

viii. Feedback

Users can send feedback, and admin can view and assess it for improvements and enhancements.

ix. Logout

The system allows the user to log out from their account.

x. Manage Bank Information

The admin can insert the bank loan information by using admin dashboard.

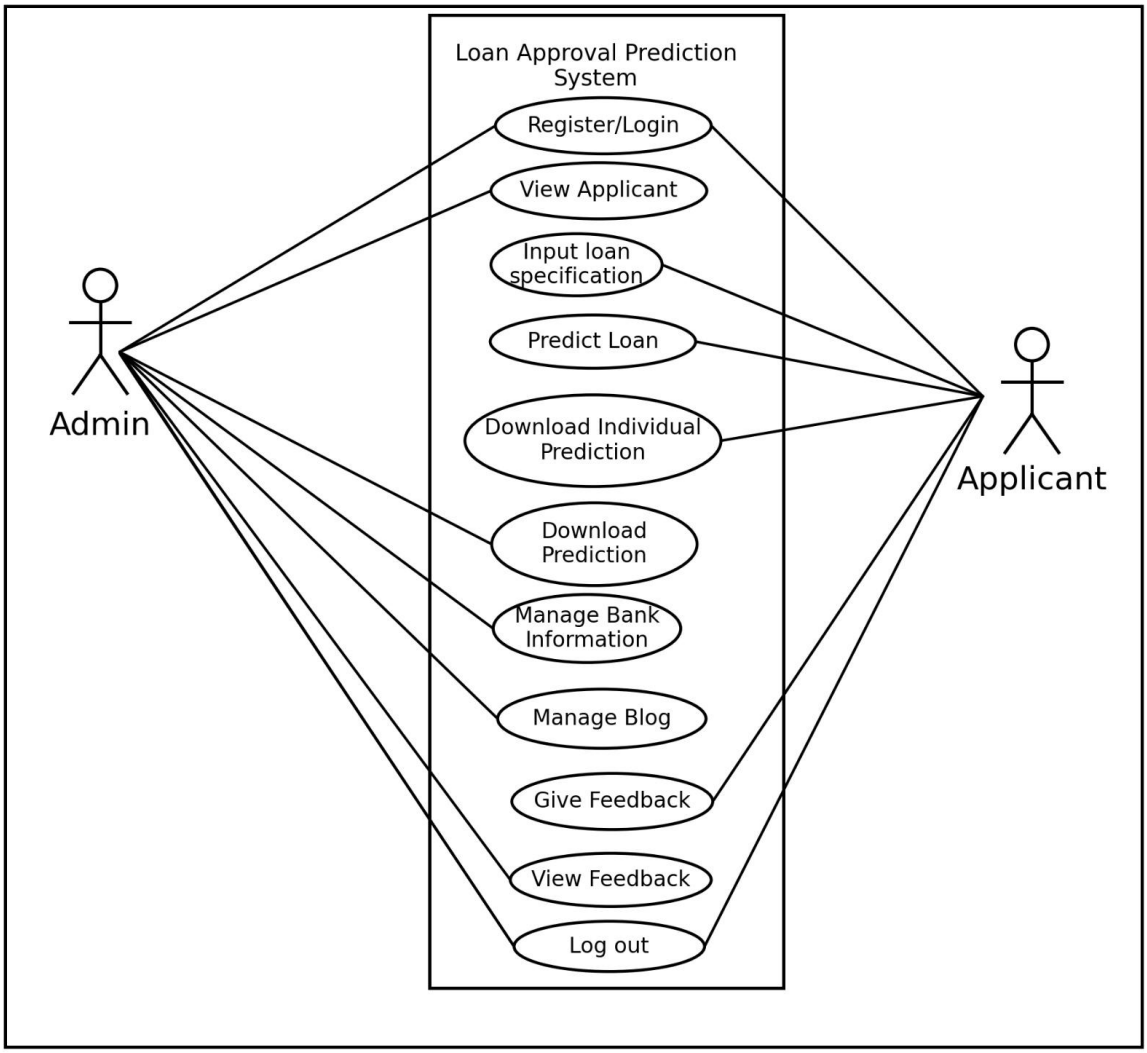


Figure 3.1 Use Case Diagram of Loan Approval Prediction System

b. Non-Functional Requirement

Non-functional requirements specify the qualities and characteristics that a loan approval prediction system should possess.

i. Performance

The system has fast response times to ensure efficient processing of loan predictions. It handles a high volume of loan information and large datasets without significant performance degradation.

ii. Availability

The system can operate 24 hours per week and 365 days a year. As long as the user, do not shut down the desktop.

iii. Usability

The user interface is intuitive, user-friendly, and easy to navigate for loan officers and other authorized users.

iv. Portability

Loan approval prediction shall run in any platform.

v. Compliance

The system complies with relevant regulatory requirements and industry standards, such as data protection regulations (e.g., GDPR) and financial industry regulations.

3.1.2. Feasibility Study

i. Technical

The technologies chosen, such as Next Js for the frontend Node Js for the backend, are widely used and well-documented. There are sample online resources, tutorials, and communities available to assist with any technical challenges that may arise during development. The machine learning random forest algorithm, being a well-established technique, can be implemented using various libraries and frameworks in Python. With the required knowledge and skills in web development, machine learning, and programming, the project can be successfully executed. So, this project is technically feasible.

ii. Operational

The project is operationally feasible. Within the timeframe the project can be developed as it is a college final year project. The required tasks of data collection, preprocessing, model training, and frontend-backend development can be accomplished efficiently. Effective project management, resource allocation, and communication among team members will be essential to ensure successful completion within the allocated time.

iii. Economic

As this is a college final year project without a specified budget, economic feasibility is not a major concern. The project can be implemented using open-source tools and resources, which are freely available, and hosting costs can be minimized by opting for low-cost or free hosting options. Since the necessary hardware and software resources

are typically available within the college or personal computers, the project can be completed without significant economic expenses.

iv. Schedule

The development process is scheduled to encompass the design phase and extend until the end of the semester, provides with approximately six months. While achieving a flawless final system within this timeframe might be challenging. But, following waterfall model with exact mindset allows us to allocate enough time to create an initial, functional version of the software. The rough schedule of the project is shown below (using Gantt chart):

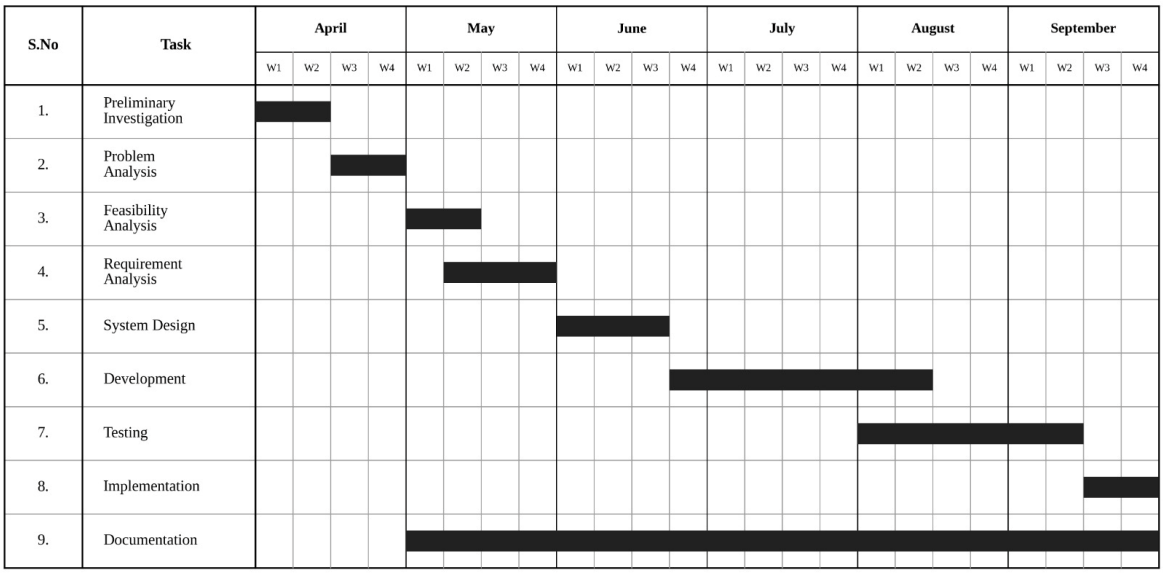


Figure 3. 2 Gantt Chart of Loan Approval Prediction System

3.1.3. Object Diagram

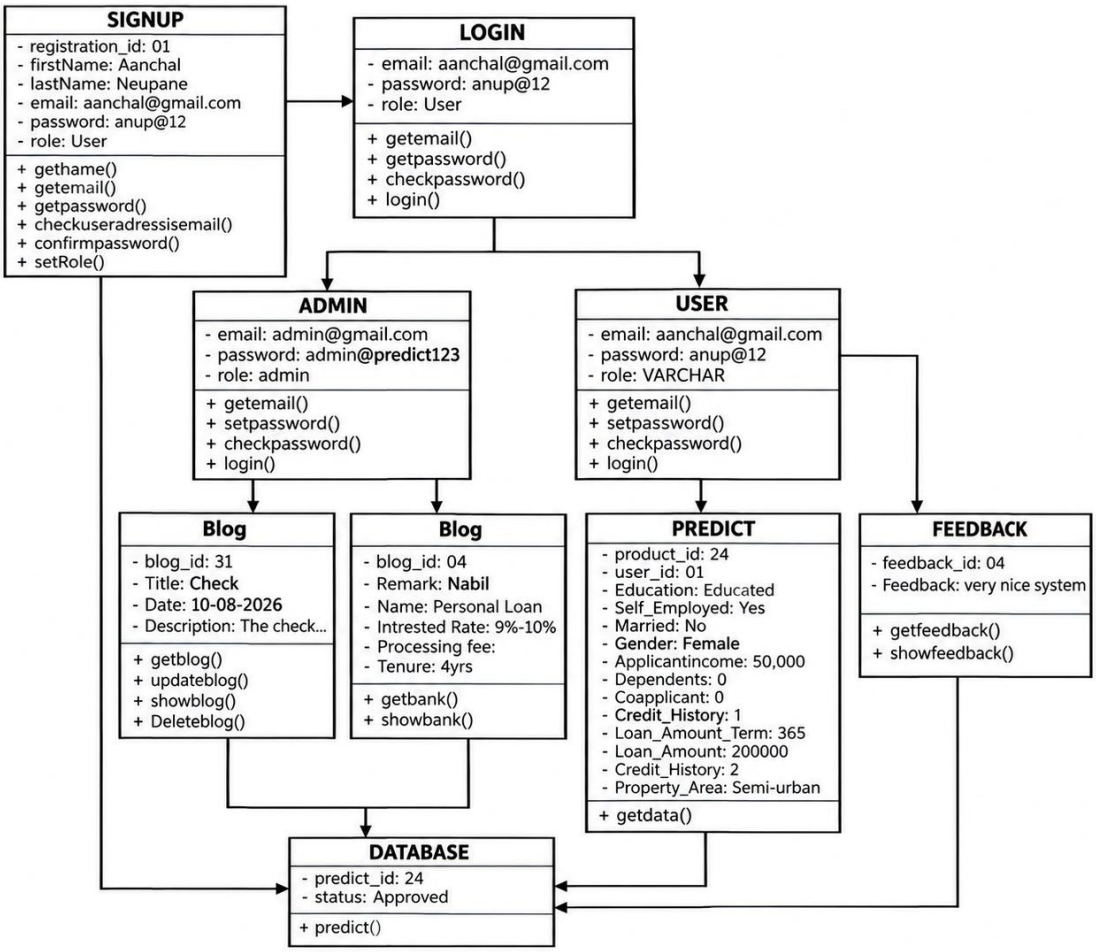


Figure 3.3 Object Diagram of Loan Approval Prediction System.

Object is an instance of a class in a particular moment in runtime that can have its own state and data values. Likewise, a static UML object diagram is an instance of a class diagram; it shows a snapshot of the detailed state of a system at a point in time, thus an object diagram encompasses objects and their relationships which may be considered a special case of a class diagram or a communication diagram.

The above object diagram shows the object and relationship of "Loan Approval Prediction System" which consists the objects: Signup, Login, User (Applicant), Admin, Blog, Feedback and Prediction.

3.1.4. Sequence Diagram

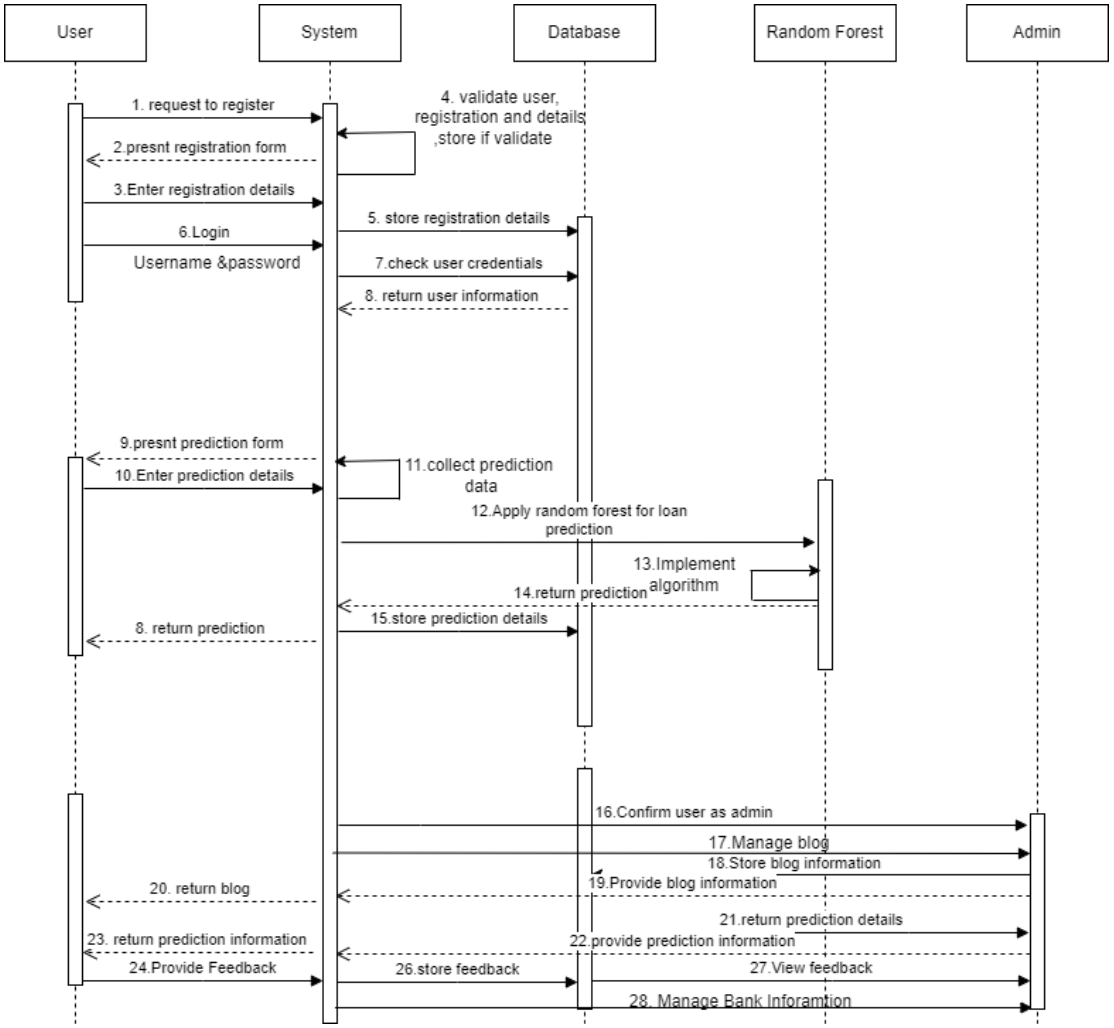


Figure 3.4 Sequence diagram of Loan Approval Prediction System

UML Sequence Diagrams are interaction diagrams that detail how operations are carried out. They capture the interaction between objects in the context of a collaboration. Sequence Diagrams are time focus and they show the order of the interaction visually by using the vertical axis of the diagram to represent time what messages are sent and when.

The given sequence diagram shows the interaction between different components of the Loan Approval Prediction System using Random Forest.

The diagram shows the interactions between the user, system, database, random forest, and admin. The user initiates the interaction by filling prediction form to find out their status about approval or disapproval of Loan.

The admin can log into the system, add blog about loan information and view all the applicant status about loan. Overall, this sequence diagram provides a high-level overview of the interactions between different components of the Loan Approval Prediction System.

3.1.5. Activity Diagram

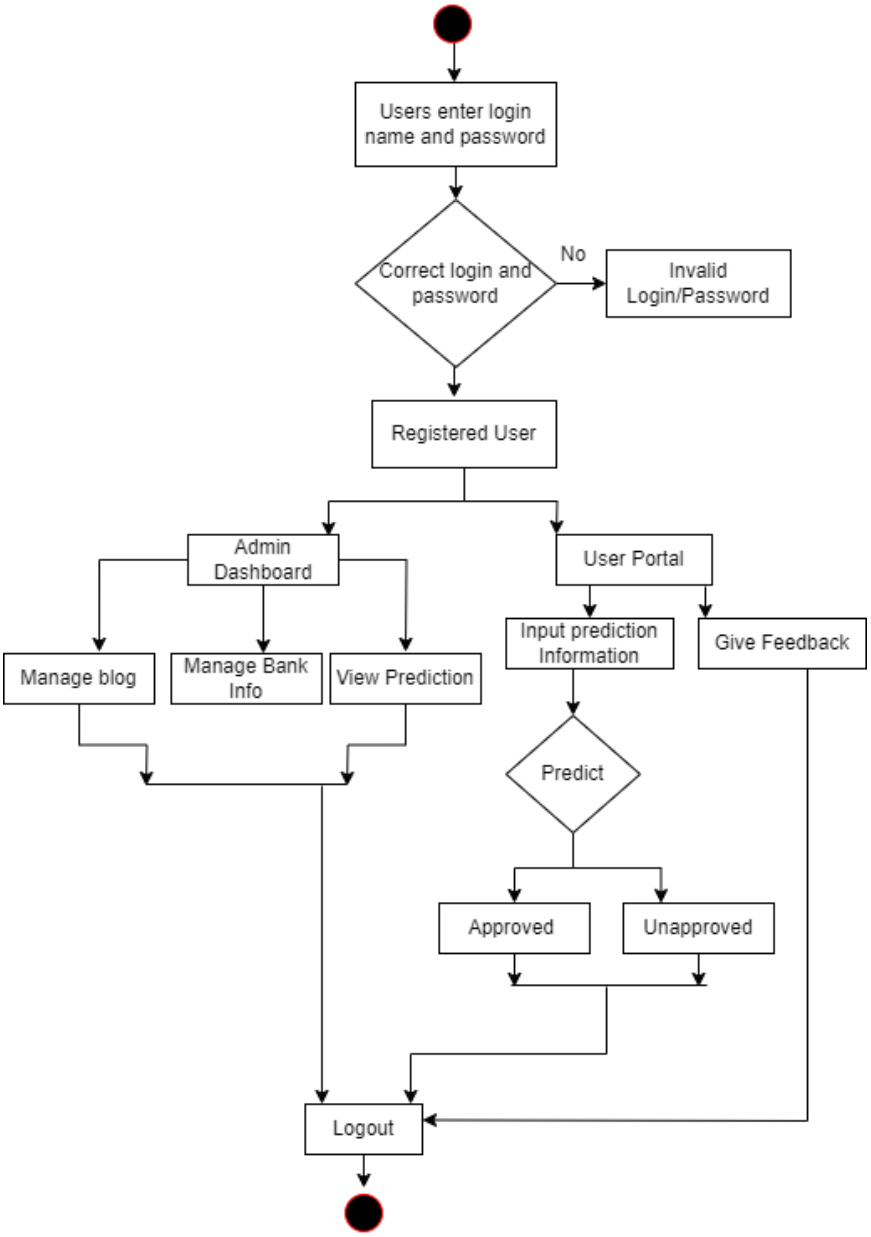


Figure 3.5 Activity diagram of Loan Approval Prediction System using Random Forest

Activity diagram is essentially an advanced version of flow chart that modeling the flow from one activity to another activity

The activity diagram starts with the Login activity. After the user logs in, the system checks if the user is registered in the database. If not, the system prompts the user to register. Once the user is registered, the system checks whether the user is an admin or a regular user. If the user is an admin, the system allows the admin to manage blog and view information of

users with status of loan approval. Once the admin completes the activities, they can log out of the system.

If the user is a regular user, the system provides prediction form where user can predict the loan approval. If the status is approved than approved message is popup and if not than unapproved message is obtained. Once the user perform prediction, they can log out of the system.

3.2. System Design

3.2.1. Component Diagram

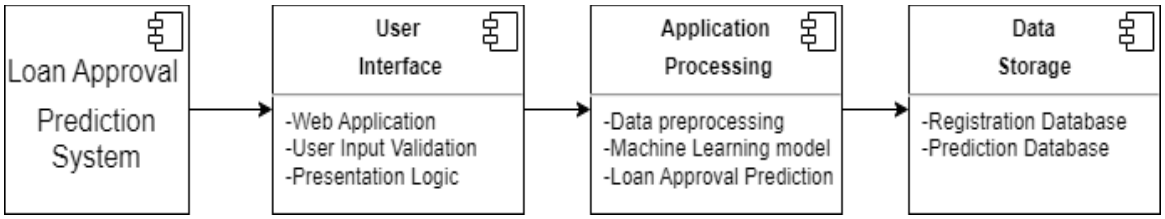


Figure 3.6 Component Diagram of Loan Approval Prediction System.

A component diagram for a loan approval prediction system visually represents the software components used in the system and shows how they interact and depend on each other. The user interface handles user interactions and includes a web application for users to submit loan applications. It also manages user input validation and presentation logic. The application processing component is responsible for processing loan applications. It includes sub-components like data preprocessing and a machine learning model for prediction. The data storage component stores various data relevant to the system, such as loan application details and user profiles. It typically includes databases to manage this information.

3.2.2. Deployment Diagram

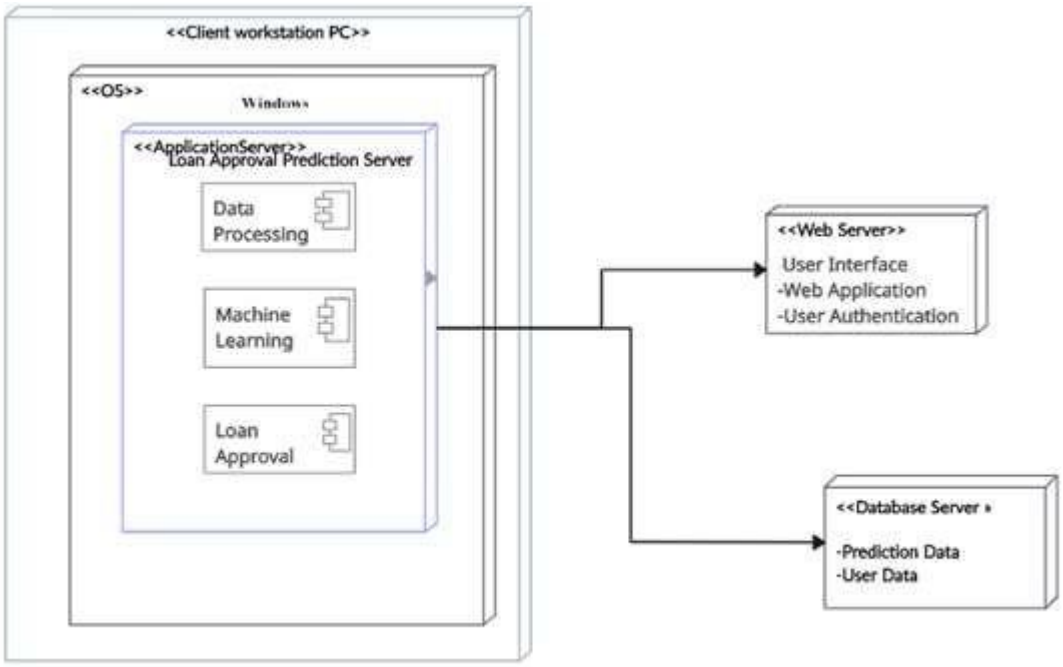


Figure 3.7 Deployment Diagram of Loan Approval Prediction System.

A deployment diagram is a visual representation in systems engineering that illustrates the physical deployment of software components and hardware nodes in a system architecture. It shows how different components of a web application work together to process a loan application.

The Client Workstation PC is the user's computer, where they can access the loan prediction web page. The web page is hosted on a Web Server, which is responsible for delivering the web page to the user's browser.

When the user submits their prediction form, the web server forwards the data to an Application Server. The application server is responsible for processing the loan prediction and making a decision on whether or not to approve it.

The application server stores the loan prediction data and user information in a Database Server. This data can be used to generate reports and track loan performance.

3.2.3. Algorithm Details

Random Forest is a popular machine learning algorithm used for both classification and regression tasks. It is an ensemble learning technique that combines the predictions of multiple decision trees to improve overall prediction accuracy and reduce overfitting. At the core of a Random Forest are decision trees. Decision trees are simple, tree-like

structures that make decisions by splitting the data based on feature values. Random Forest can handle missing values in the dataset without much preprocessing.

Step 1: The class is initialized with three parameters: n_estimators (number of decisions trees in the ensemble), max_depth (maximum depth of each decision tree), and random_state (seed for random number generation).

Step 2: An empty list, estimators, is created to store the trained decision trees.

Step 3: The fit method is used to train the random forest classifier on the input data X and corresponding labels y. It iterates n_estimators times to create multiple decision trees.

Step 4: np.random.choice is use to ensure diversity among the trees inside fit method.

Step 5: DecisionTreeClassifier is created inside fit method with the specified max_depth and random_state.

Step 6: Train the decision tree on the selected features by calling the fit method with the appropriate subset of the input data X and labels y.

Step 7: Append the trained decision tree, along with the selected features, to the estimators list, forming the ensemble of decision trees for the random forest classifier inside fit method.

Step 8: Create an predict method with empty array, predictions, to store the predictions from each decision tree in the ensemble.

Step 9: Iterate over each decision tree in estimators in predict method.

Step 10: Obtain the final predictions by applying majority voting along each row of the predictions array.

Step 11: Use np.apply_along_axis and np.argmax(np.bincount(x)) to select the most frequent class label in predicts method.

Step 12: Return the majority vote predictions.

Chapter 4. Implementation and Testing

4.1. Implementation

4.1.1. Tools Used

No special hardware requirements were necessary to develop the loan approval prediction system using model based on Random Forest algorithm. The system was built and tested on a 64-bit computer running Windows. This configuration was sufficient to handle the data processing and model training tasks required for the project.

The various software tools used to develop this project are as follows:

a. CASE tools

CASE tools used in Loan Approval Prediction System:

i. Figma

Figma is used by us to design wireframes for the initial case. Icons, logos, and assets can be easily created using Figma.

ii. Visual Studio Code

Visual Studio Code is used by us because it is a code editor that is employed for redefining and optimizing the development and debugging of modern web and cloud applications. Instant productivity is achieved by us through features such as syntax highlighting, bracket-matching, auto-indentation, box-selection, snippets, and more provided by VS Code.

iii. Google colab

Google Collaboratory, is a free online platform that provides a way to run and write Python code using Jupyter notebooks. It's useful for tasks like data analysis, machine learning, and collaborative coding because it offers a cloud-based environment, allowing users to access powerful computing resources and share their work with others. It's particularly handy for those who don't want to set up their own coding environment and need access to GPUs for computationally intensive tasks.

b. Programming Languages

i. JavaScript:

JavaScript is a lightweight, interpreted programming language. It is open and cross platform. Integration of HTML and CSS can be easily achieved using JavaScript. Next JS and Node JS, the JavaScript frameworks, are utilized by us. Next.js is a popular React framework for constructing server-rendered web applications. Node.js is a server-side JavaScript run-time environment capable of executing asynchronous I/O through its event-driven architecture.

ii. Python:

Python is a versatile and beginner-friendly programming language renowned for its simplicity and readability. It is widely employed in various domains, encompassing web development, data analysis, artificial intelligence, and beyond. The clear syntax of Python facilitates code writing and comprehension, rendering it an excellent selection for individuals ranging from novices to seasoned developers. Its extensive libraries and a sizable community bolster its widespread adoption. In the realm of computation and analysis, specific Python libraries are required to conduct analytics. Essential packages like SKlearn, NumPy, Pandas, Matplotlib, and the Flask framework are utilized for this purpose.

iii. Flask:

It is a lightweight WSGI web application framework. It is designed to make getting started quick and easy, with the ability to scale up to complex applications.

c. Database Platforms

MySQL is a popular open-source relational database management system (RDBMS) that allows to store, manage, and retrieve data efficiently. It's commonly used for applications that require structured data, like websites and business software. MySQL uses a structured query language (SQL) to interact with the database, making it easy to work with tables and perform various data operations. It's reliable, fast, and widely supported, making it a preferred choice for many database-driven applications.

4.1.2 Implementation Details of Module

a. Admin Module

The Admin Module is an important component of our system that provides administrative privileges and control over various aspects of the application. Here are the implementation details:

i. View User Predictions:

The admin has the ability to access and view all predictions made by users within the system. This feature allows the admin to gain insights into the trends and patterns of user predictions.

ii. Manage Blog Posts:

Within the admin module, there is a functionality that enables the admin to manage the content of the blog section. They can upload new blog posts to share information or updates, delete outdated posts, and update existing ones to keep the content relevant and informative.

iii. Manage Bank Information:

The system administrator can input bank loan details via the admin dashboard, which will then be displayed on the loan details page of the website.

iv. View feedback:

The administrator has the ability to access and review feedback provided by system users. This information serves as valuable input for potential modifications and improvements to the system in the future**.**

b. User Module

The User Module serves as a platform for authorized users to make predictions about loan approvals and retrieve their historical prediction records. Here are the details:

i. Authorized User Login:

To access the system's prediction capabilities, users must log in with their authorized credentials. This login process ensures that only authenticated individuals can use the prediction service.

ii. View Past Predictions:

Users also have the ability to access and review their past loan approval predictions. This feature allows users to track the accuracy of their previous predictions and assess their financial decisions over time.

iii. Give Feedback

Users of the system have the opportunity to provide feedback based on their experiences within the system. This feedback mechanism allows them to share their thoughts, opinions, and observations about their interactions with the system. They can use this feature to offer insights, suggestions, or comments to help enhance and improve the system for both themselves and other users.

4.2. Testing

Software Testing is a method to check whether the actual software product matches expected requirements and to ensure that software product is defect free. It involves execution of system components using manual or automated tools to evaluate one or more properties of interest. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements. The following things were done during the process of "Loan Approval Prediction System" testing:

- Tests were planned before testing begun.
- The entire tests were prepared as per users' requirements.
- Analytical tools were used to develop test cases.
- A testing strategy was adopted and applied.

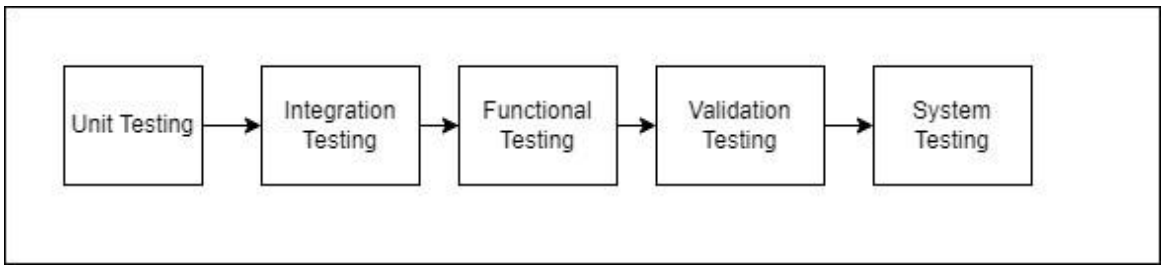


Figure 4.1 Testing of Loan Approval Prediction System

The two main Test approach of Software Engineering are:

1. White box testing

White box testing involves testing the internal structure and logic of software, ensuring that all code paths, logical decisions, loop boundaries, and internal data structures are tested to meet specific requirements and maintain validity.

2. Black box Testing

Black box testing centers around evaluating software against its functional requirements, utilizing diverse input scenarios to detect issues like missing functions, interface errors, data structure problems, performance issues It also initialized and terminate errors, with the ultimate aim of improving software quality by identifying and resolving these concerns.

4.2.1. Test cases for Unit Testing

Unit testing is done to validate that each unit of the software code performs as expected. It is done during the coding phase of an application by the developers. This test is done to determine the working of the individual modules. Loan Approval Prediction System includes the various modules that are tested at the development processes.

Table 4.1. Test case for Login

Test case id	1		
Test case description	User Login		
Prerequisites	User must be registered Enter the valid username and password. Click login		
Test scenario 1	User enter a wrong username		
Test data	Username: abc@gmail.com Password: student		
Step	Expected output	Actual Result	Pass/fail
1	Incorrect information	Username or Password is Invalid	pass
Test scenario 2	User enter a right password		

Test data	Username: example@gmail.com Password: abcd		
Step	Expected output	Actual Result	Pass/fail
1	Correct information	Username or Password is Invalid	Pass
Test scenario 3	User enter all details successfully		
Test data	Username: example@gmail.com Password: example1		
Step	Expected output	Actual Result	Pass/fail
1	User account login	Login Successful	Pass

Table 4.2. Test case for admin login

Test Case ID	2		
Test Description	Login as admin		
Prerequisites:	1.Valid id and password		
Test Scenario	1. Enter the valid id and password. 2. Click login		
Test Data	User name: admin@gmail.com Password: admin@prediction123		
Step	Expected Result	Actual Result	Pass/Fail
1.	Logged in.	Login Successful.	Pass

Table 4.3. Test case for blog management by admin

Test Case ID	3
Test Description	Add the blog
Prerequisites:	All the field must be filled.

Test Scenario	1.Enter the title and description 2. Click submit button. 3.Click update button. 4.Update the title. 5.Click Delete button.		
Test Data	Title: Loan rate of Nabil Bank Ltd. Description: From Shrawan 24, Nabil Bank Ltd has decided…		
Step	Expected Result	Actual Result	Pass/Fail
1. 2. 4.	Blog Added and Published. Title Updated Delete Success	Blog Added and Published. Title Updated Delete Success	Pass Pass Pass

4.2.2. Test case for system testing

Since this test concentrates on each unit of the software as implemented in source code, the test was conducted keeping every small module in consideration. The following is the report of a few bugs that were overcome during the test and those remain unchecked.

Table 4.4 Test case for system testing.

S.N.	Bugs:	Solution:
1	Lack of input validation leading to unexpected inputs. **Error description:** Invalid Input.	Validate the input data in proper way.
2	System predict that an applicant will be approved for a loan, but they are actually denied. **Error description:** Incorrect predictions.	Train the model with error free data, select appropriate feature and avoid overfitting.

3	The system crashes when a user tries to submit a loan application with incomplete information. **Error description:** Unexpected result.	Handle incomplete form properly. Notify the user about the information that is missing.

Most of the error is caused by human because human develop system and manipulation is done by human themselves only, hence nobody can deny the possibility of having committing some or other mistake or the existence of bugs. Even though, extreme condition should be taken care for maintaining the data and respective entries of the system for the testing of report. A set of test materials is nothing more than a list of possible problems in a program and a set of procedure for determining whether the problems actually exist and are significant or not.

4.3. Result Analysis

The result analysis of the loan approval prediction system involves assessing the accuracy and performance of the predictions made by the system. This analysis typically includes evaluating metrics such as precision, recall, F1 score, and accuracy to determine how well the model is performing in classifying loan prediction as approved or unapproved.

Confusion Matrix:

The confusion matrix is a powerful tool for assessing the performance of classification models. It can help us to understand the strengths and weaknesses of our models and to identify areas where they need to be improved. Various performance metrics, including accuracy, precision, recall, and F1 score, can be computed using the confusion matrix. The confusion matrix of the Loan Approval Prediction System is shown below:



Figure 4.2. Confusion Matrix of Loan Approval Prediction System.

Accuracy:

The accuracy of the classifier is the percentage of loan applications that it correctly predicted.

Mathematically,

Accuracy = (TP + TN) / (TP + TN + FP + FN) =0.825

An accuracy of 0.825 means that a machine learning model is correctly predicting the target value 82.5% of the time. It means the model has average performance for performing prediction.

Precision:

The precision of the classifier is the percentage of loan applications that it predicted to be approved that were actually approved.

Mathematically,

Precision = TP / (TP + FP) = 0.84530

A precision of 0.84530 means that, of all the instances that the model predicted to be positive, its correct about 84.53% of the time. The precision value shows that the model is relatively good at predicting true positive.

Recall:

The recall of the classifier is the percentage of loan applications that were actually approved that the classifier predicted to be approved.

Mathematically,

Recall = TP / (TP + FN) =0.916167.

A recall of 0.91616 means that the model correctly identified 91.616% of all the positive instances. The model is quite effective at identifying most of the true positive cases.

F1 score:

The F1 score is a harmonic mean of precision and recall. It is a good metric to use when both precision and recall are important.

Mathematically,

F1 score = 2 * (Precision * Recall) / (Precision + Recall) = 0.87931.

An F1 score of 0.87931 means that the model is performing well overall. It is able to identify both true positives and true negatives with a high degree of accuracy.

Receiver operating curve:

The Area Under the Curve (AUC) is a critical metric in machine learning that quantifies the performance of a binary classification model. It plots the true positive rate (TPR) against the false positive rate (FPR) at different classification thresholds. A higher AUC indicates a better classifier. A perfect classifier would have an AUC of 1.0, while a random classifier would have an AUC of 0.5.

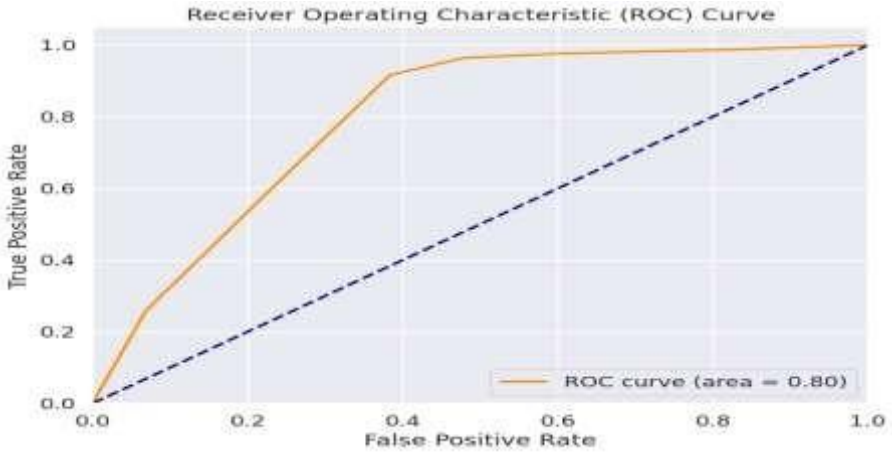


Figure 4. 3 ROC curve of Loan Approval Prediction System

An AUC of 0.80 indicates that it is able to distinguish between positive and negative instances with 80% accuracy. This is a good performance, but it is not perfect. There is still room for improvement, and the classifier could be made more accurate by using more data, training for longer, or using a different machine learning algorithm.

Chapter 5. Conclusion and Future Recommendation

5.1. Conclusion

In conclusion, a project that incorporates Random Forest for Loan Approval Prediction is developed. Loan Approval Prediction System uses machine learning to predict the approval or disapproval of loan. It uses data management to store and organize the data of users, blog and prediction information. Furthermore, the prediction model was integrated with Flask, a lightweight Python web framework, to improve the user experience and offer a user-friendly interface for system interaction. The system allows the administrator to manage the blog post and view user with loan status by providing easy to use interface. Due to the power of machine learning, loan approval prediction system successfully predicts loan approval outcomes with a good degree of accuracy.

5.2. Future Recommendation

There are many different machine learning algorithms like linear regression, support vector machine, K-nearest neighbor and more that can be used for loan approval prediction. The experiment can be performed with different algorithms to see which one performs best on dataset and can implement the algorithm with higher accuracy. Another way to further improvement is to use more data and feature to train model to improve accuracy. The ensemble learning can be implemented which is the technique that combines the predictions of multiple machine learning models to produce a more accurate prediction. The system can consider both individual user information and financial and risk assessment parameters to make accurate and compliant predictions about loan approval. This helps banks to streamline the loan approval process and make better decisions about which loans to approve.

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Appendices

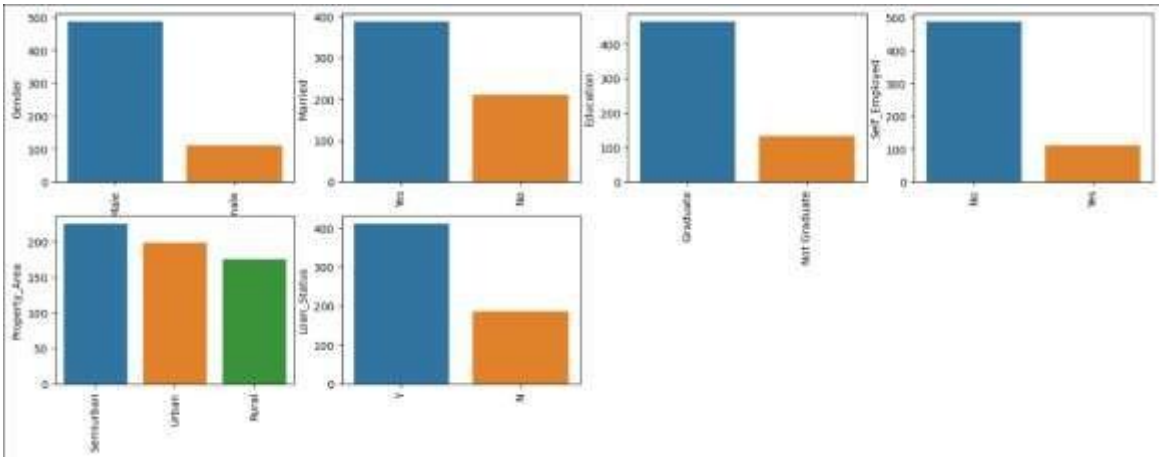


Figure 1: Bar Diagram of Loan Prediction Model

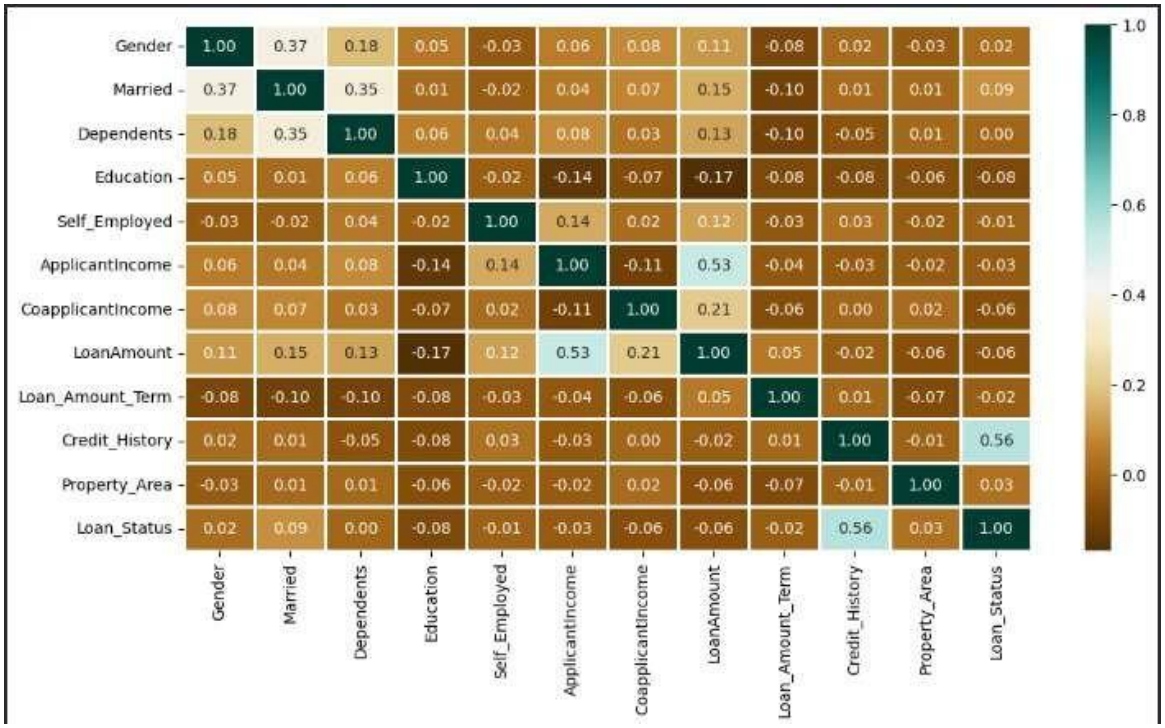


Figure 2: Heat Map of Loan Prediction Model

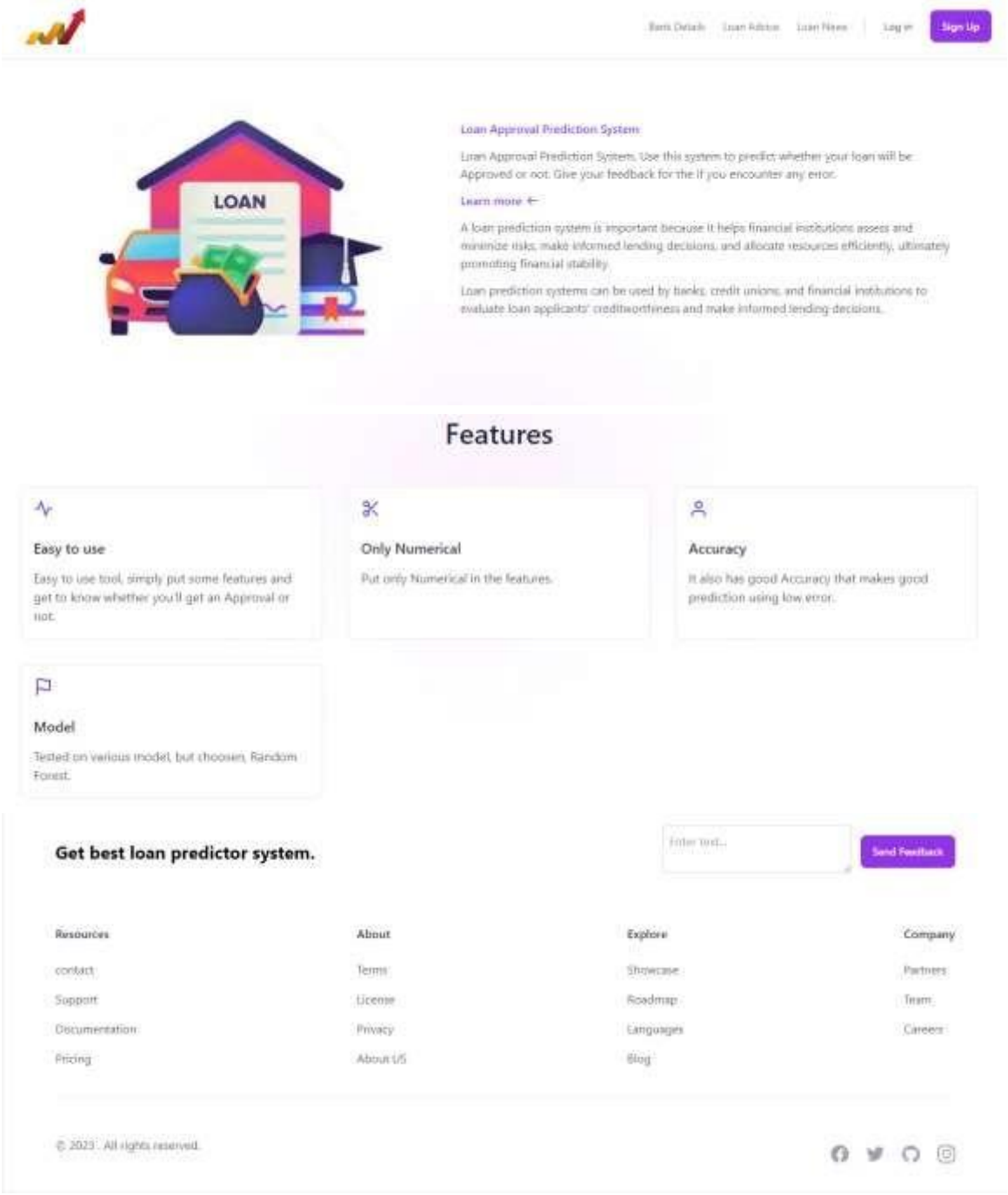


Figure 3: Homepage of Loan Prediction System

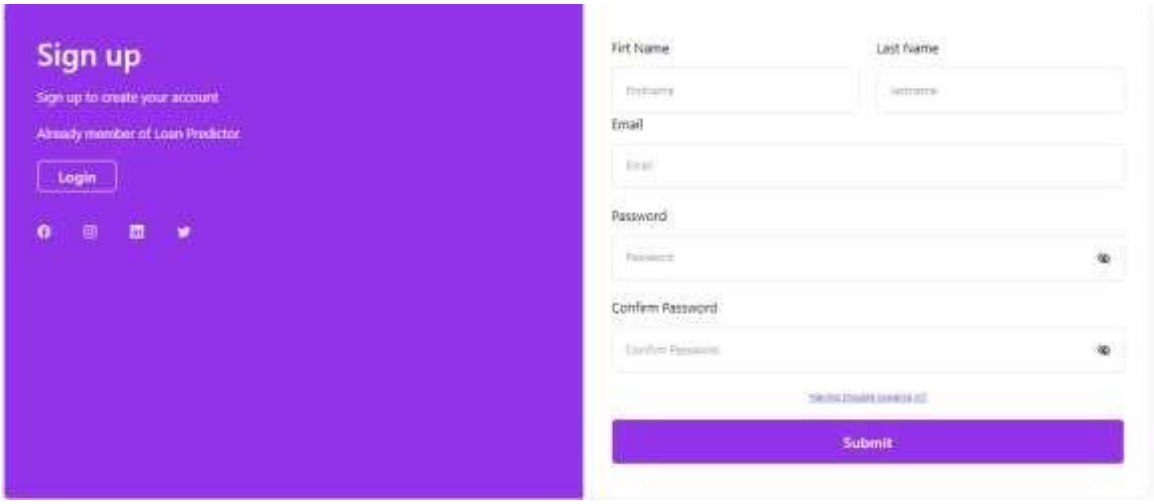


Figure 4: Signup Form of Loan Prediction System



Figure 5: Login Form of Loan Prediction System



Figure 6: Navbar after user Login

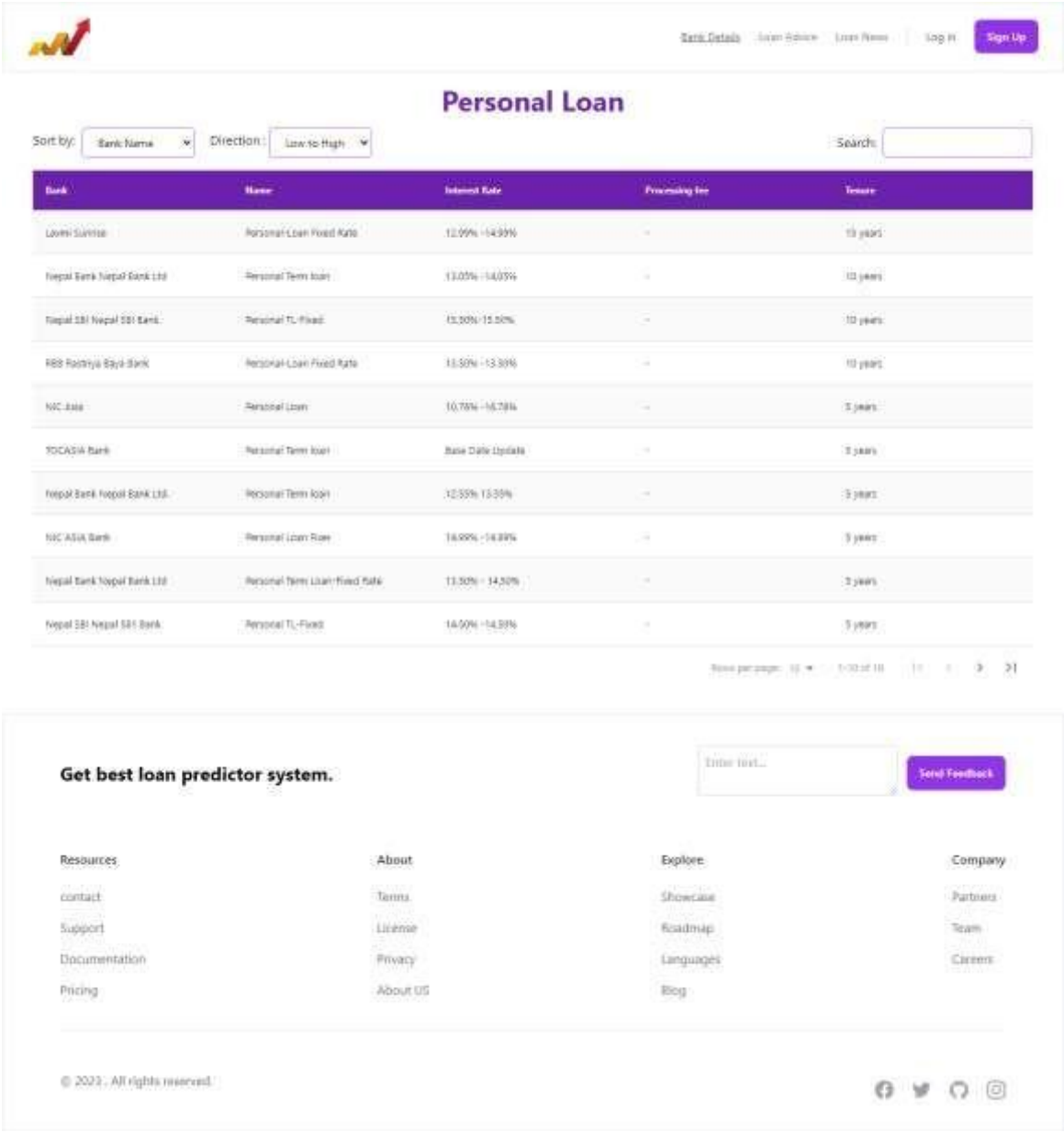


Figure 7: Loan Details Page

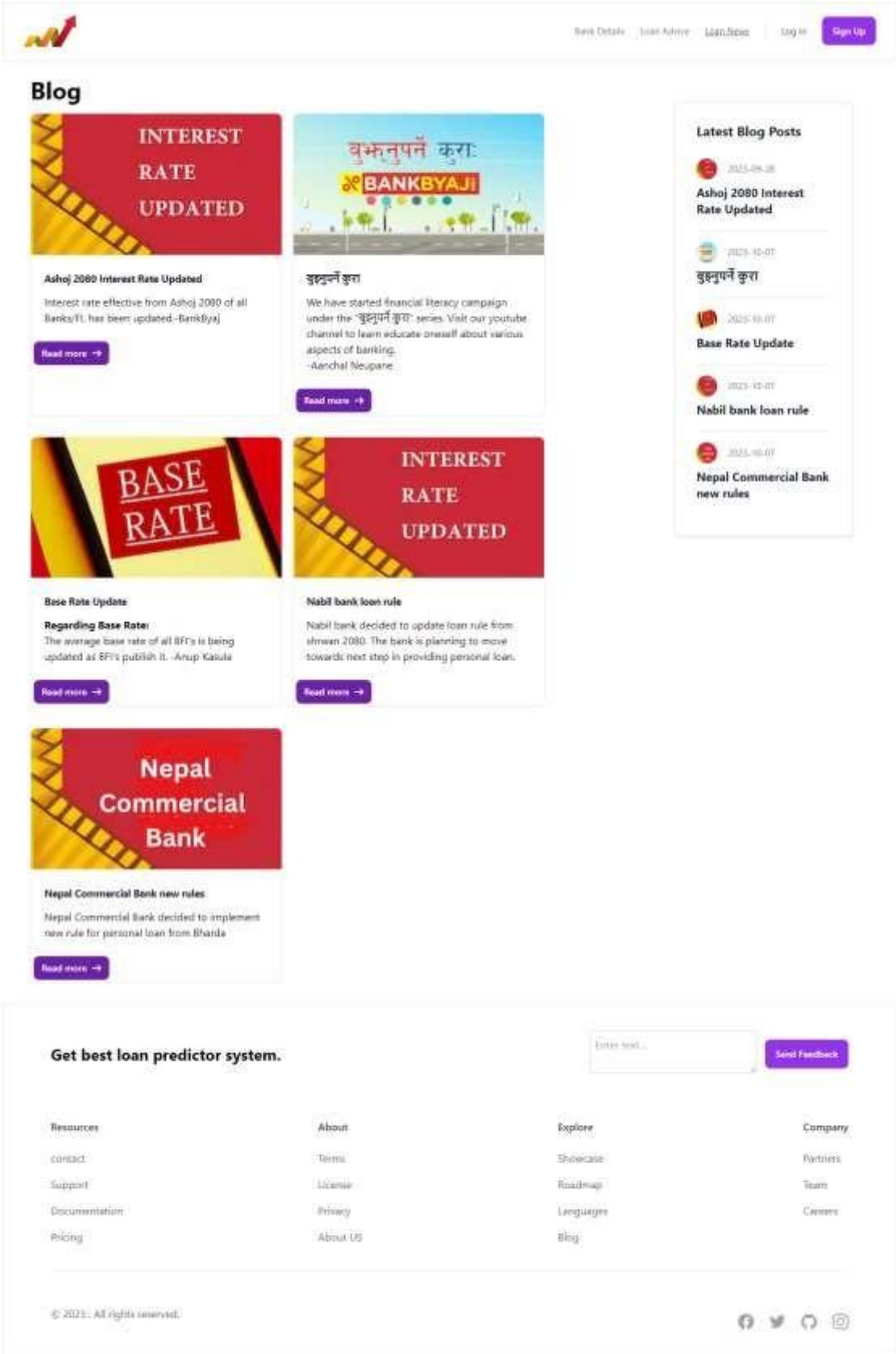


Figure 8: Blog Page

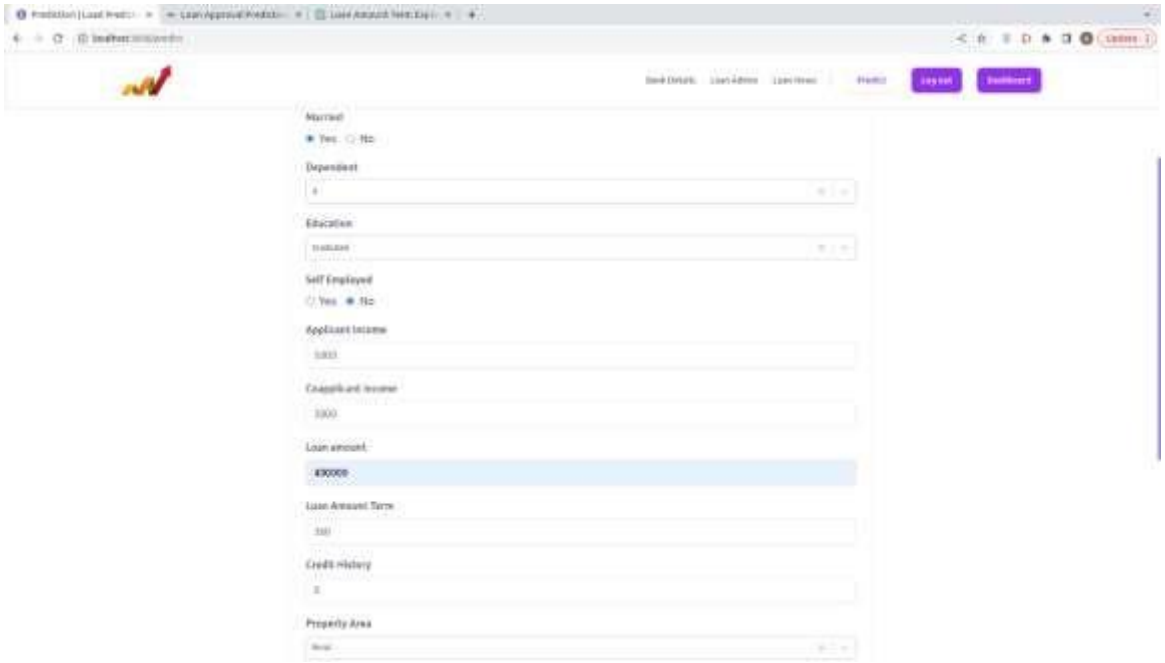


Figure 9: Loan Prediction Form

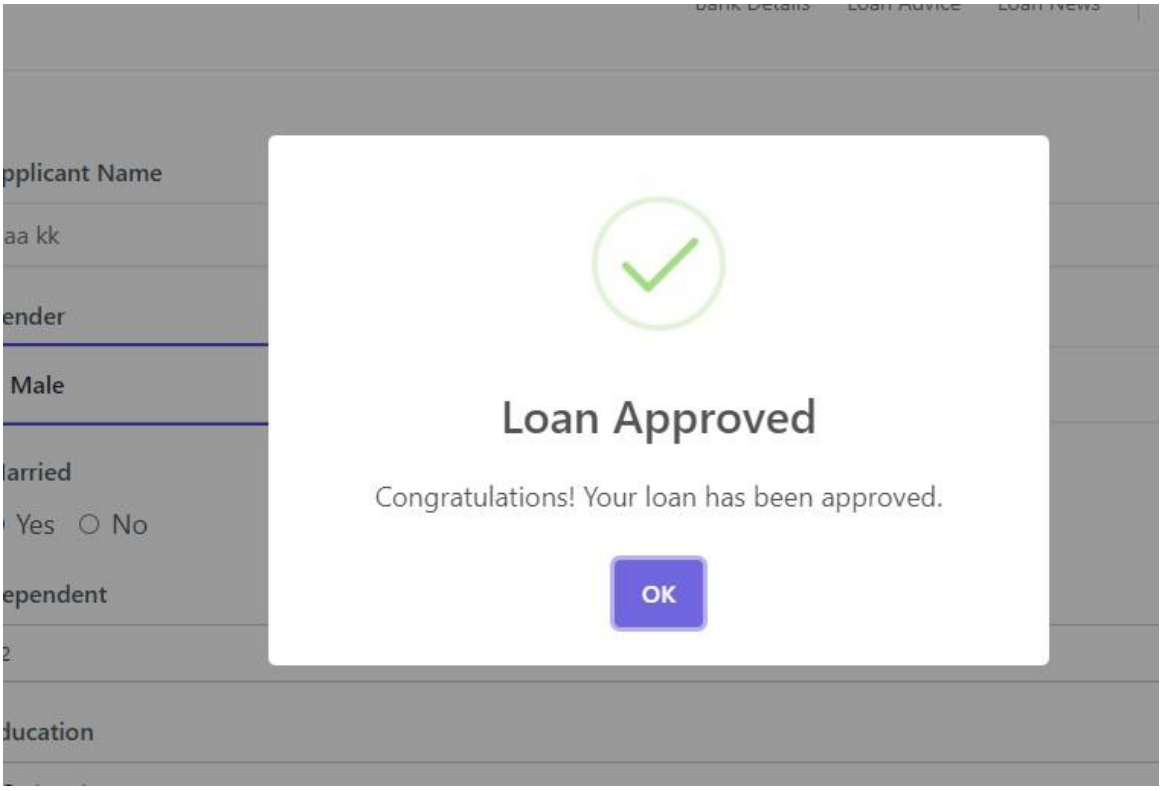


Figure 10: Loan Approved Message Popup

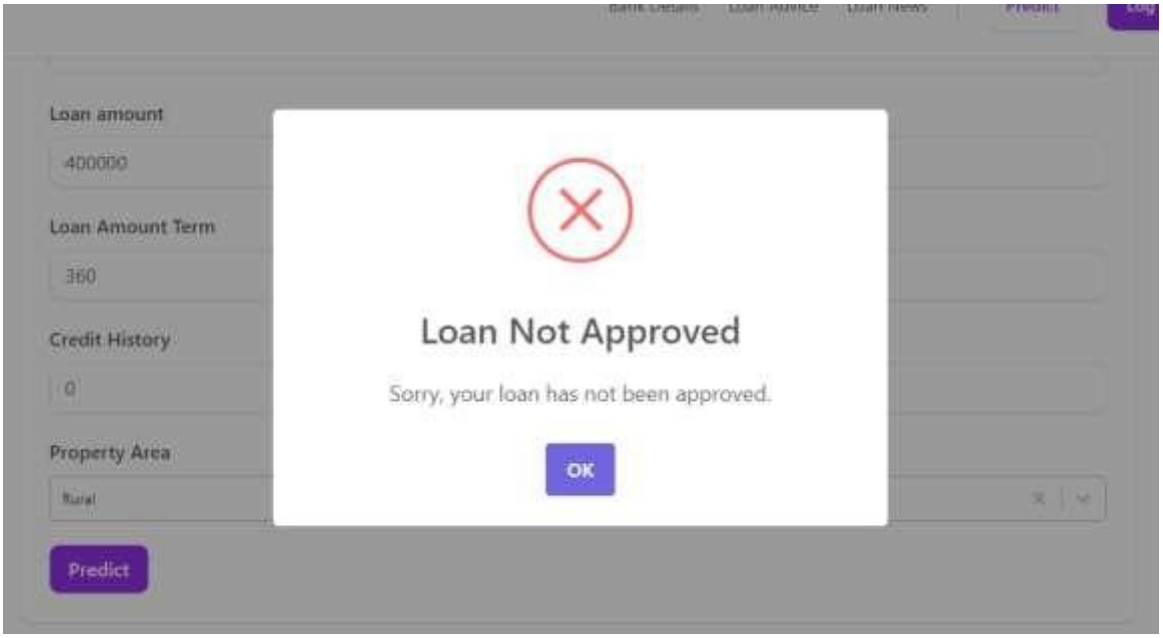


Figure 11: Loan Not Approved Message Popup

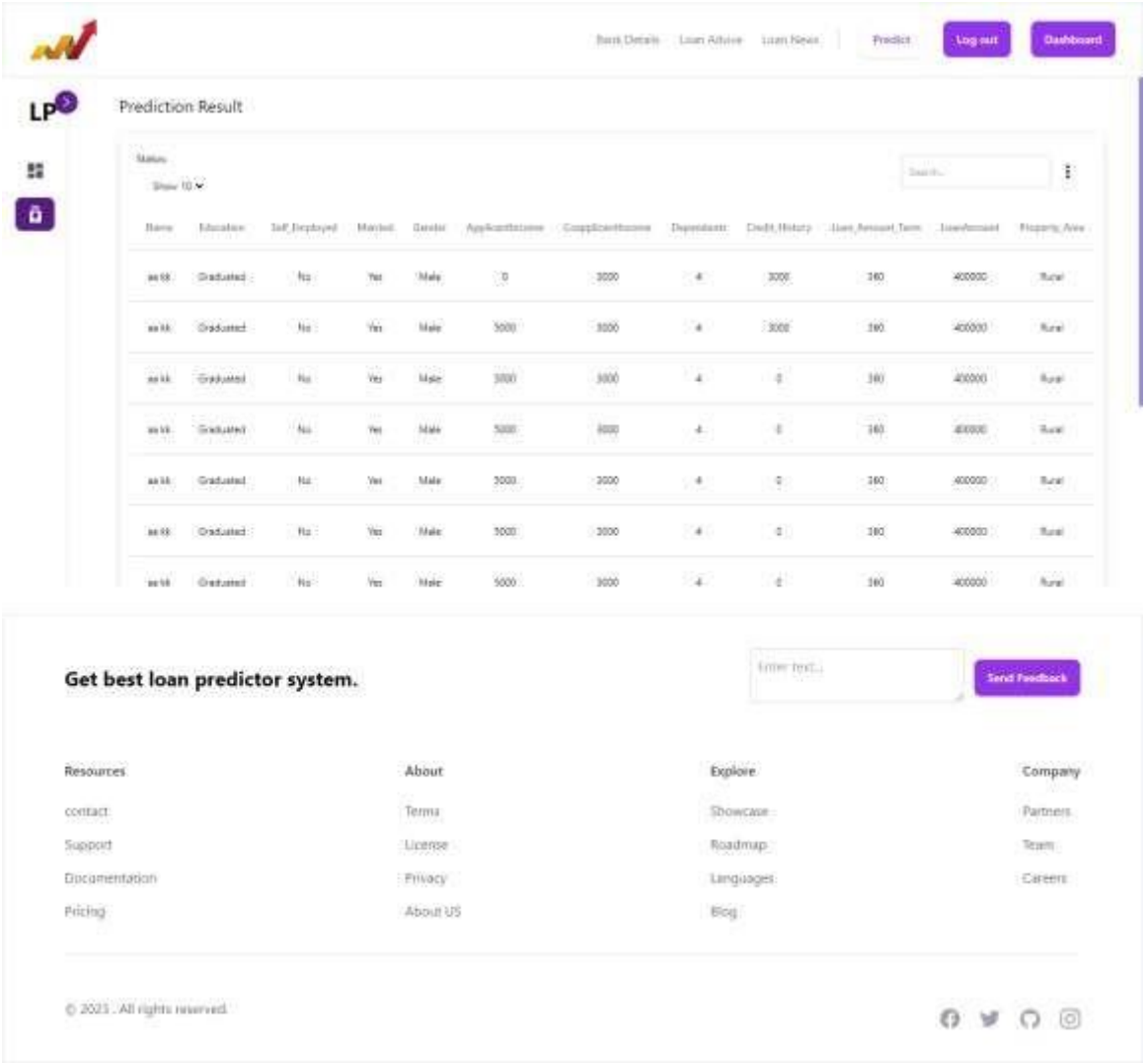


Figure 12: User Prediction Results

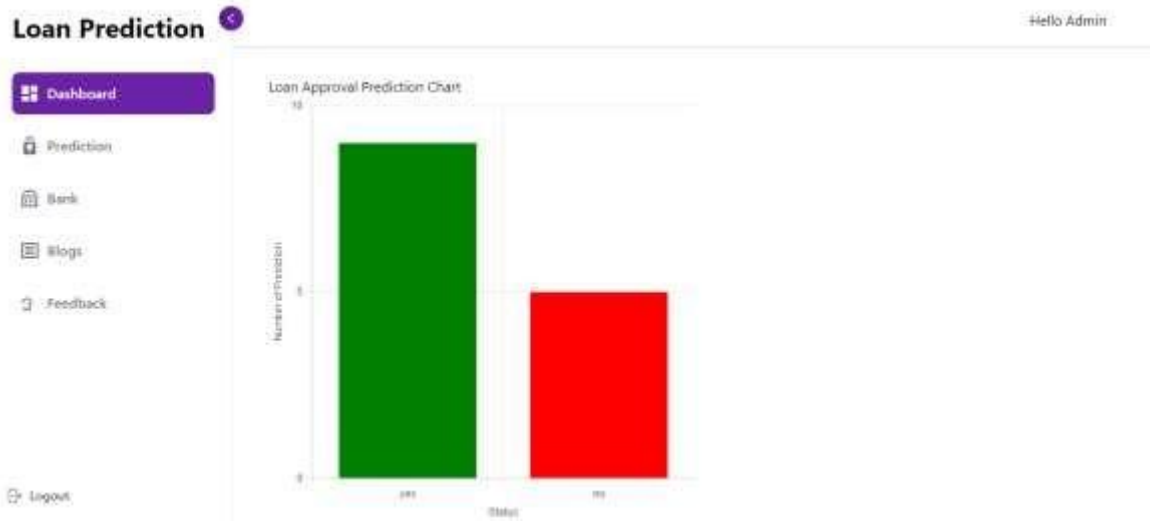


Figure 13: Admin Dashboard of Loan Prediction System

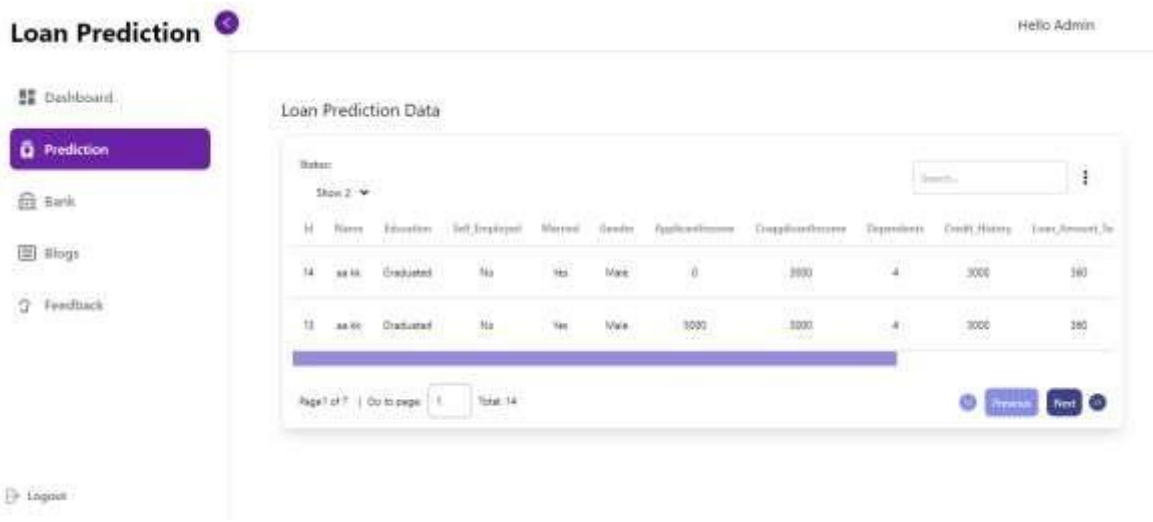


Figure 14: All user Prediction Result in Admin

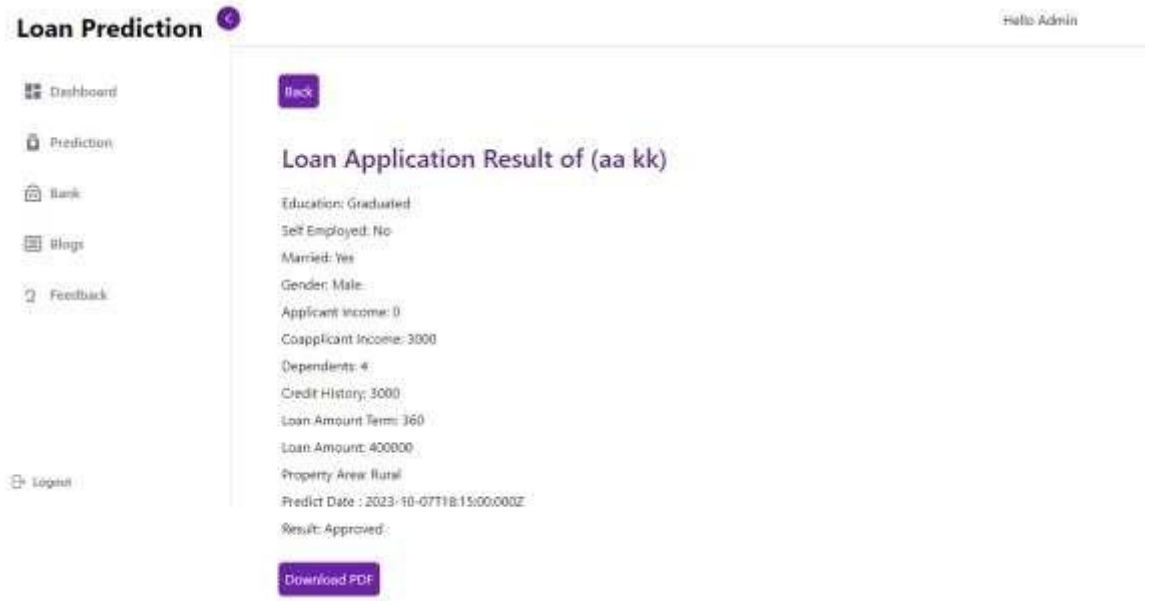


Figure 15: Single User Loan result

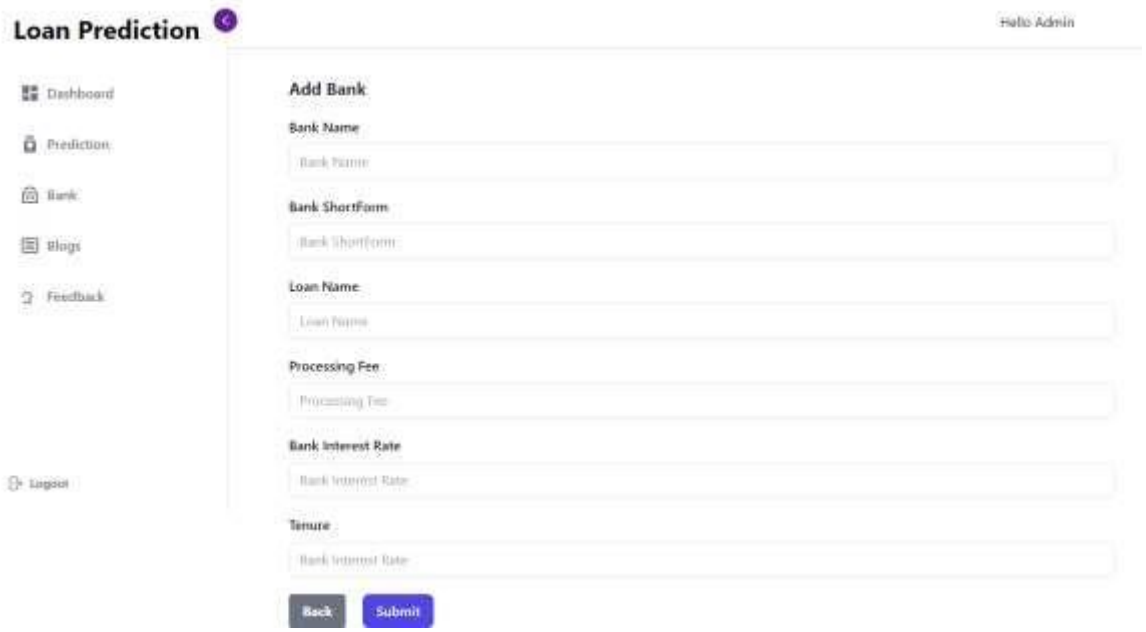


Figure 16: Add Bank and Loan Details of Bank by Admin

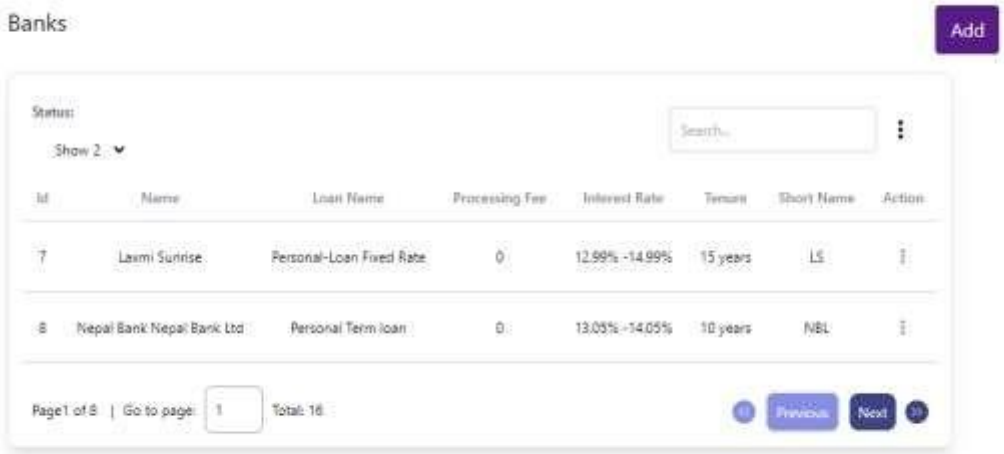


Figure 17: Bank Details view in admin panel

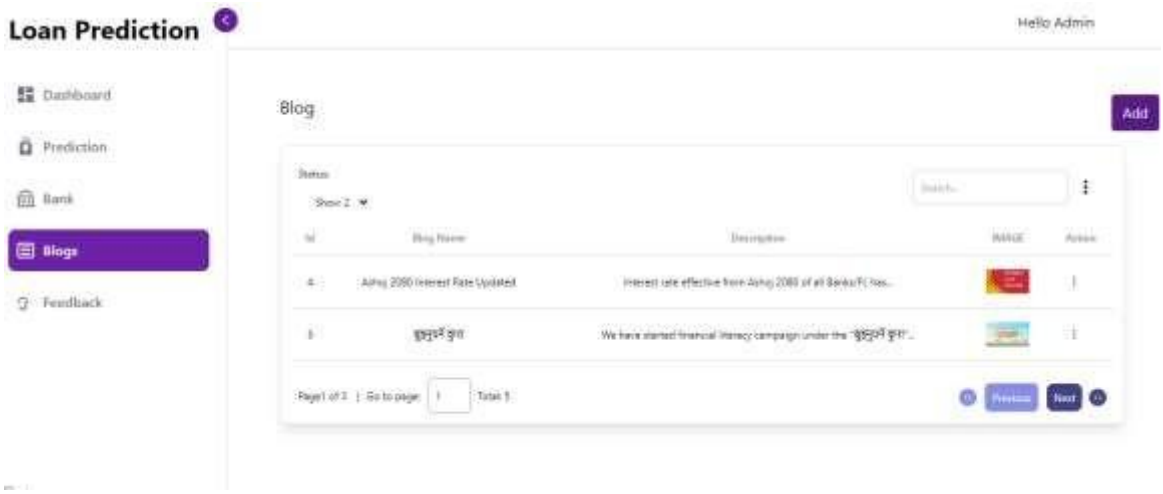


Figure 18: Blogs data view in admin panel

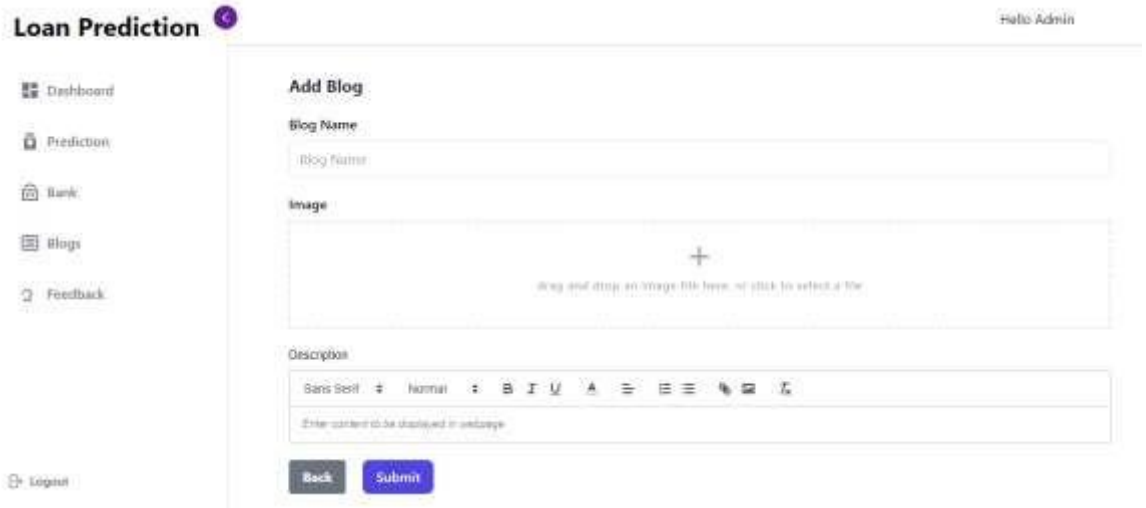


Figure 19: Add Blogs form



Figure 20: User feedback